

First single- λ super-rated 200G DD PON downstream with 340-ps/nm CD tolerance enabled by dual-SSB transmission

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Abstract We experimentally demonstrate 224-Gbit/s/ λ downstream transmission in a direct-detection PON system enabled by transmission of a dual single-sideband signal and individual reception of each sideband. We demonstrate a loss budget exceeding 33 dB and a chromatic-dispersion tolerance beyond 340 ps/nm. ©2026 The Author(s)

Introduction

The ITU-T SG15/Q2 study group is evaluating technology candidates for Passive Optical Networks (PONs) beyond 50 Gb/s in the Very High Speed PON (VHSP) project (G.sup88), targeting service rates of at least 100 Gb/s per wavelength and 200 Gb/s aggregated [1]. Reaching such service capacities requires even higher line rates to compensate for the PON protocol overhead – a concept referred to as ‘super-rating’. In deployment scenarios where GPON co-existence is strictly required by operators, VHSP downstream (DS) cannot be operated close to the zero-dispersion wavelength of the fiber, which substantially increases the amount of chromatic dispersion (CD) to be tolerated by the system. While the adoption of coherent transceivers in PON is explored by ITU-T and would allow achieving high spectral efficiency and CD tolerance, such transceivers are not favourable in terms of implementation complexity, power consumption, and cost. Recently, hybrid PON schemes have gained attraction as a compromise offering enhanced CD tolerance compared to traditional intensity-modulation (IM) and direct detection (DD), while keeping hardware complexity at a moderate level. As an example, multiplexing of multiple wavelength channels each employing single-sideband (SSB) modulation and DD has been demonstrated [2,3], yet wavelength multiplexing comes at the cost of an increased component count.

In this paper, we demonstrate a dual-SSB VHSP DS scheme, where the upper and lower modulation sideband (USB and LSB) of a single optical carrier carry independent information [4-5]. This approach allows transmission and DD of 200 Gb/s service rate on a single wavelength with more than 33 dB loss budget, thereby improving spectral efficiency and reducing the number of hardware components compared to conventional SSB transmission, while preserving its CD tolerance. We further combine the dual-SSB

technique with digital CD compensation at the transmitter or receiver to compensate the nonlinear phase response added by the dispersive fiber, which enhances the maximum system reach further. Our proof-of-concept demonstration confirms the feasibility of CD tolerances beyond 340 ps/nm for future VHSP systems and supports full coexistence with existing PON technologies.

Dual-SSB PON Concept

Figure 1(a) illustrates the proposed PON system. At the optical line terminal (OLT), a dual-SSB signal is generated with a dual-drive Mach-Zehnder modulator (DD-MZM) that is fed by a single laser source. Two independent data streams are combined and applied to one arm of the DD-MZM ($d_1 + d_2$), while the other arm is driven by the difference of the two associated Hilbert transforms ($\mathcal{H}\{d_1\} - \mathcal{H}\{d_2\}$) [6]. At the receiver side, the dual-SSB signal is pre-amplified, then the two SSB signals are separated with an optical filter [5-6], and finally each SSB signal is individually detected with a PIN photodiode with subsequent transimpedance amplifier (PIN-TIA). Note that the dual-SSB approach offers additional flexibility because simple ONUs could only receive a single sideband (100 Gb/s), see Fig. 1(b), whereas premium ONUs could receive both sidebands (200 Gb/s), see Fig. 1(c).

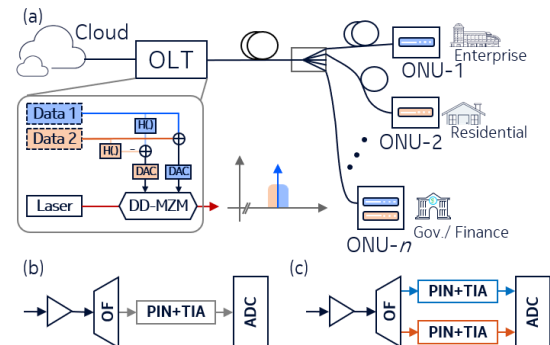


Fig. 1: Proposed dual-SSB PON system. (a) Single-laser dual-SSB transmitter based on a DD-MZM and Hilbert filtering. (b-c) Candidate Rx architectures for 100 Gb/s and 200 Gb/s employing optical filtering and direct detection.

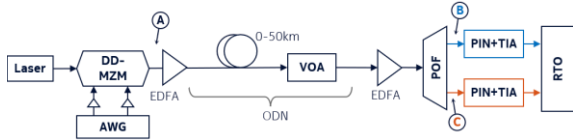


Fig. 2: Experimental setup for proof-of-concept demo.

Experimental Setup

To test the viability of the proposed dual-SSB DS PON scheme, we set up a proof-of-concept experiment using benchtop components, see Fig. 2. At the OLT, the transmitter employs a micro-integrated tunable laser source (μ TLS) with a linewidth of 100 kHz and with an output power of 8 dBm. The optical carrier is modulated in a 40-GHz DD-MZM. Two independent 112 Gb/s NRZ data streams are generated with 2^{15} symbols each, up-sampled and the frequency-dependent attenuation of the transmitter electronics is pre-compensated. One DD-MZM arm is then driven by the sum of the two data streams, while the other arm is driven by the difference of their Hilbert-transformed signals. The Hilbert transform is implemented by convolution with a 64-tap finite impulse response filter [7]. The two resulting waveforms are fed to an arbitrary waveform generator (AWG) with a sampling rate of 224 GSa/s and with an analog 3-dB bandwidth of 55 GHz.

Figure 3 shows optical spectra that have been measured for different transmitter settings at position ① of Fig. 2: (a) shows a conventional double-sideband (DSB) signal as obtained for intensity modulation as a reference; (b) presents the spectra of individually generated SSB signals. Sideband suppression ratios of 18 dB and 16 dB are achieved for the LSB and USB, respectively; (c) shows the spectrum of the final dual-SSB signal, where the USB and LSB carry independent information. The modulated DS signal is subsequently amplified to 16 dBm using an EDFA to compensate for modulation and transmitter insertion losses. The optical distribution network is emulated using 0–50 km of standard single-mode fiber (SSMF), followed by a variable optical attenuator (VOA) to emulate PON splitter losses. At the ONU, an EDFA with 25 dB gain serves as a pre-amplifier. Subsequently, a programmable optical filter (POF) is used to separate the USB and LSB channels while simultaneously suppressing the amplified spontaneous emission (ASE) noise introduced by the optical amplifiers prior to DD. The POF is programmed to emulate a 100- Note that a considerable portion of the EDFA gain is required to compensate the high insertion loss (7 dB) of the POF. We expect that the required pre-amplifier gain can be reduced with a dedicated application-specific filter design. The measured spectra at positions ② and ③ are shown in Fig. 3(d), where the EDFA noise floor is clearly visible. We remark that a future VHSP system will most likely employ semiconductor

optical amplifiers (SOAs) instead of EDFAs, which will come with an additional performance penalty. Each SSB signal is individually detected using a PIN-TIA, followed by a 256-GSa/s real-time oscilloscope (RTO) with 42 GHz analog bandwidth. Finally, the bit-error-rate (BER) performance is evaluated after offline digital signal processing (DSP) of each individual SSB signal, including timing recovery, a T-spaced 40-tap feed-forward equalizer (FFE), and a 4-symbol memory maximum-likelihood sequence estimation (MLSE) [8–10]. The MLSE is primarily required to mitigate the bandwidth limitations of the RTO and could likely be omitted or replaced by a decision feed-back equalizer in implementations using more broadband converters. 2x2 MIMO processing of the two SSB signals can potentially reduce crosstalk and improve performance in the future [6].

Experimental Results

We have conducted a series of experiments to evaluate the proposed dual-SSB system at an overall line rate of 224 Gbit/s. Figure 4(a) shows exemplary electrical spectra received after 10-km of SSMF, illustrating that DD of SSB modulation prevents CD-induced power fading caused by destructive interference of signal-mixing terms. As a first experiment, we investigated the BER performance in a back-to-back (B2B) configuration as a function of the received optical power (ROP), see Fig. 4(b). The non-filled and filled markers in the figure represent the single-SSB (112-Gbit/s) and dual-SSB (224-Gbit/s) cases, respectively, whereas the grey markers show the performance of DSB modulation as a reference. As seen, the single-SSB performance is slightly worse compared to DSB, which we attribute to signal-signal mixing. In contrast, a power penalty of ~ 6 dB is observed when moving from single-SSB to dual-SSB transmission. This power difference is also visible in the optical spectra shown in Fig. 3(b) and (c), where the sideband power is 6 dB weaker for the dual-SSB case. This effect is caused by the fact that the drive voltage amplitude is kept the same, while a sum of two data

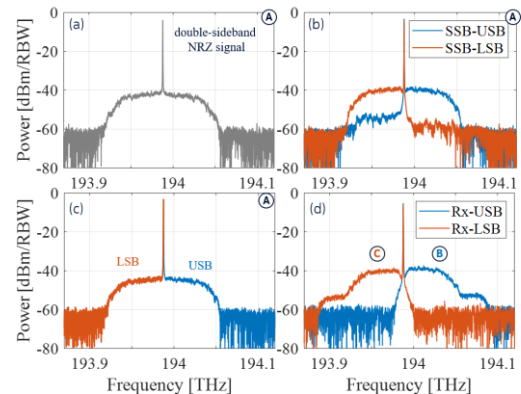


Fig. 3: Measured optical spectra at different points in the experimental setup. RBW: 150 MHz.

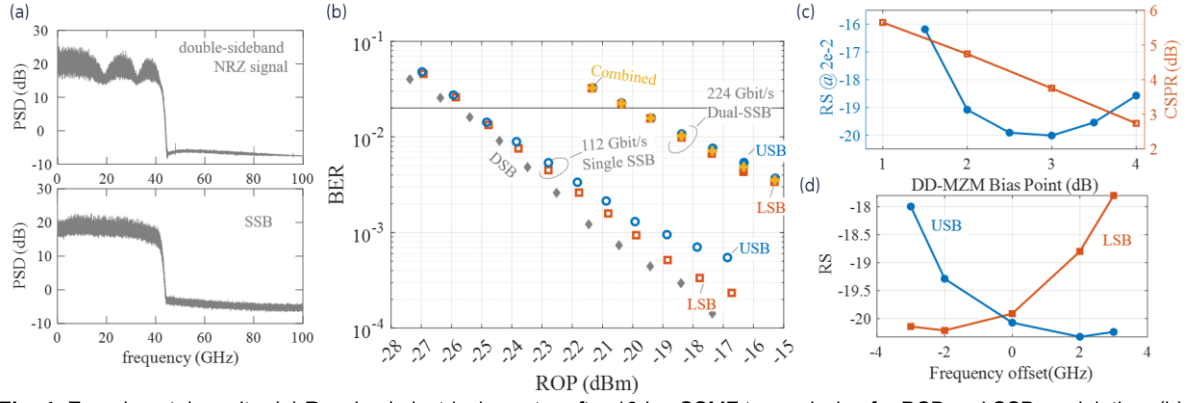


Fig. 4: Experimental results. (a) Received electrical spectra after 10-km SSMF transmission for DSB and SSB modulation. (b) BER as a function of received optical power (ROP) in B2B configuration for single-SSB and dual-SSB transmission. (c) DD-MZM bias optimization. (d) Receiver sensitivity versus frequency offset between the laser and optical filter.

streams is used for the drive, which effectively leads to a halved electrical RMS level of each tributary relative to the single-SSB case. Part of this penalty can be recovered by increasing the RMS levels and by clipping the signal peaks, at the expense of slightly reduced CD tolerance. Figure 4(c) shows the receiver sensitivity (RS) at the pre-FEC BER threshold of 2×10^{-2} and the corresponding carrier-to-signal power ratio (CSPR) as a function of the DD-MZM bias. A bias point of ~ 3 dB provides the optimum RS at a CSPR of ~ 4 dB and was used for all experiments discussed in this paper.

Next, we investigate the wavelength-alignment tolerance between the laser and the receiver optical filter, as shown in Fig. 4(d). We observe a performance variation of approximately 2 dB for a wavelength offset of ± 3 GHz. Finally, we have performed dual-SSB fiber transmission and measured the PON loss budget as a function of accumulated dispersion at 224 Gbit/s, see Fig. 5. To compensate the nonlinear phase response of the dispersive fiber, we combined the dual-SSB transmission with different digital CD pre- or post-compensation techniques. The blue curve represents direct dual-SSB reception and shows a CD tolerance up to 170 ps/nm with ~ 3.5 dB sensitivity penalty. Adding Kramers–

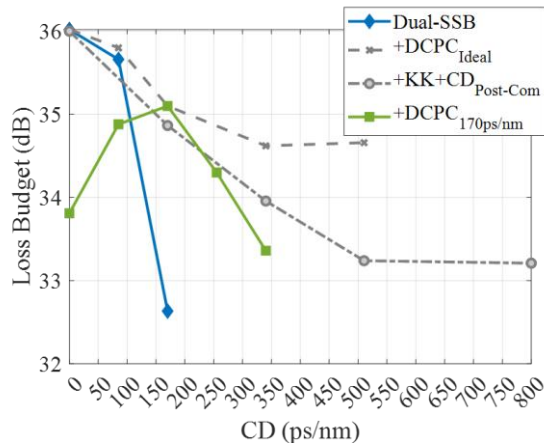


Fig. 5: Achieved ODN loss budgets as a function of accumulated CD for different transmitter & receiver DSP.

Kronig (KK) field reconstruction at ~ 2.3 samples per symbol and subsequent digital CD post compensation, resulted in a moderate dispersion penalty up to 800 ps/nm, see the grey dash-dotted curve in Fig. 5. However, KK reception introduces additional DSP complexity at the ONU, which is typically undesired. Furthermore, the limited adoption of KK receivers in other high-volume market segments may hinder its practicality for cost-sensitive VHSP deployments. The dashed grey curve with cross markers corresponds to transmitter-side digital CD pre-compensation (DCPC) applied for accumulated CD values of 85, 170, 340, and 510 ps/nm. DCPC has been implemented by convolution of two additional 128-tap finite impulse response filters for each arm of the DD-MZM [11]. In this case, we achieve a loss budget exceeding 34 dB with less than 1.5 dB sensitivity penalty even for more than 500 ps/nm of dispersion. However, individual optimization of DCPC per OLT-ONU link requires link-specific CD information and burst-like downstream transmission (potentially with ONU grouping) [12], which ultimately increases system complexity. Finally, we have investigated combining dual-SSB with a static DCPC filter used for all OLT-ONU links assuming a dispersion level of 170 ps/nm. Using this configuration, we achieve a loss budget exceeding 33 dB while the sensitivity variation across the wide CD range from 0 to 340 ps/nm remains below 2.8 dB.

Conclusions

We have demonstrated a 224-Gbit/s/λ DD-PON system enabled by dual-SSB transmission and individual reception of each SSB signal via experimental results. By employing static transmitter-side CD pre-compensation, the system achieves 0–340 ps/nm CD tolerance without additional ONU complexity while doubling the spectral efficiency compared to a conventional single-SSB scheme.

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Speed PON," 51st European Conference on Optical Communication (ECOC), Copenhagen, Denmark, 2025.
DOI: 10.1109/ECOC66593.2025.11262947

References

- [1] ITU-T, "G.Sup88 : Point to multipoint passive optical access system requirements and transmission technologies above 50 Gbit/s per wavelength", Oct 2025
https://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-G.Sup88-202510-1!!PDF-E&type=items
- [2] R. Koma et al., "Demonstration of C-band, 50-Gbit/s $\times$ 4 λ single-sideband NRZ-signal transmission through 40-km SMF using 25G-class APD and simple feed-forward equalizer for direct-detection-based 50G-TWDM-PON," in Proc. Eur. Conf. Opt. Commun. (ECOC), 2025.
DOI: 10.1109/ECOC66593.2025.11263220.
- [3] Y. Liu et al., "WDM 100G-PON over C-band 20–50 km SSMF with linear equalizer and 44 dB power budget," IEEE Photon. Technol. Lett., 2026.
DOI: 10.1109/LPT.2026.3654631.
- [4] H.-C. Chien, Z. Jia, J. Zhang, Z. Dong, and J. Yu, "Optical independent-sideband modulation for bandwidth-economic coherent transmission," Opt. Express, vol. 22, pp. 9465–9470, 2014.
DOI: 10.1364/OE.22.009465.
- [5] Y. Zhu, X. Ruan, K. Zou, and F. Zhang, "Beyond 200G direct-detection transmission with Nyquist asymmetric twin-SSB signal at C-band," J. Lightwave Technol., vol. 35, no. 17, pp. 3629–3636, Sept. 2017.
DOI: 10.1109/JLT.2017.2718098.
- [6] L. Zhang, T. Zuo, Q. Zhang, J. Zhou, E. Zhou and G. N. Liu, "Single Wavelength 248-Gb/s transmission over 80-km SMF Based on Twin-SSB-DMT and Direct Detection," 42nd European Conference on Optical Communication (ECOC), Dusseldorf, Germany, 2016
- [7] C. Füllner et al., "Complexity Analysis of the Kramers–Kronig Receiver," in *Journal of Lightwave Technology*, vol. 37, no. 17, pp. 4295–4307, 1 Sept. 1, 2019, doi: 10.1109/JLT.2019.2923249
- [8] W. Lanneer and Y. Lefevre, "Machine Learning-Based Pre-Equalizers for Maximum Likelihood Sequence Estimation in High-Speed PONs," 2023 31st European Signal Processing Conference (EUSIPCO), Helsinki, Finland, 2023, pp. 1425–1429,
DOI: 10.23919/EUSIPCO58844.2023.10289760
- [9] C. Füllner et al., "First 100G NRZ-OOK PON Demonstration With >31 dB Loss Budget and Coexistence Study Over Field-Deployed Fiber," 2025 Optical Fiber Communications Conference and Exhibition (OFC), San-Francisco, USA, 2025, W1F.2
- [10] M. Verplaetse et al. "Super-Rated IM/DD PON Downstream Demonstration at 100G Net Rate Using Line Rates up to 124 Gb/s." 51st European Conference on Optical Communication (ECOC), Copenhagen, Denmark
DOI: 10.1109/ECOC66593.2025.11263040
- [11] M. Chagnon, "Optical Communications for Short Reach," in *Journal of Lightwave Technology*, vol. 37, no. 8, pp. 1779–1797, 15 April 15, 2019, doi: 10.1109/JLT.2019.2901201.
- [12] G. Rizzelli et al., "Multi-user Chromatic Dispersion DSP-based Precompensation and DD Receiver for Very High